USN | | | | | | | | | | | [24MTEVT103]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY - 2025
ELECTRIC DRIVE TRAIN AND CONTROL

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Draw and explain various power flow control modes for an ICE dominated series-parallel hybrid system.	co1	L2	10
	b.	Compare hybrid vehicles with conventional vehicles in terms of fuel economy under different driving cycles.	co2	L3	10
2.	**a.**	List the different components hence explain configuration of an Electric Vehicles.	co1	L2	10
	b.	Illustrate the general EV configuration and explain the functions of its subsystems.	co2	L2	10
3.	**a.**	Explain the working and the performance characteristics of BLDC drives.	co3	L2	10
	b.	Compare BLDC drive with Induction motor drive for HEV and EV.	co4	L2	10
		OR			
4.	**a.**	With the help of neat sketches explain the general configuration of constant V/F control of induction motors.	co4	L2	10

Jain (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

	b.	i. Justify why brushed DC motor not preferred for EV? ii. Differentiate between electric and hybrid vehicles.	co3	L2	10
5.	a.	Briefly explain the construction and working of a switched reluctance motor used in Electric vehicles.	co5	L2	10
	b.	Using the appropriate block diagram, describe the direct torque control method for a switched reluctance motor.	co5	L2	10
		OR			
6.	a.	Describe the construction and working principle of a permanent magnet synchronous motor for EV.	co5	L2	10
	b.	Discuss in detail about the field oriented control of permanent magnet motor drives	co5	L2	10
7.	a.	Explain the terms continuous rating, Intermittent overload operation, peak overload operation related to electric machines used for HEV. How are these relevant to the selection/sizing of the propulsion motor in HEV?	co6	L2	10
	b.	i. What are the fundamental sizing constraints to be considered while sizing an electrical machine for an EV? ii. Suggest the methods for improving the motor efficiency	co6	L2	10
		OR			
8.	a.	Illustrate the design of power train components(Motor, Power converter, storage devices) in Toyota Prius Electric Vehicle.	co6	L4	10
	b.	With the suitable diagram explain characteristics of variable speed electric motor drives.	co6	L2	10